API Documentation

AI-Powered Construction Site Intelligence Platform

Version 1.1 | Base URL: /api/v1 | Format: JSON

Global Conventions

Base URL: /api/v1/

Auth Header: Authorization: Bearer <access_token>

Content-Type: application/json (multipart/form-data for file uploads)

Timestamps: All timestamps in ISO 8601 UTC format: 2024-07-15T08:30:00Z

Standard Error Schema:

```
{
  "error": {
    "code": "STRING_CODE",
    "message": "Human-readable description",
    "status": 400
  }
}
```

HTTP Status	Meaning
200	OK — successful GET or action
201	Created — successful POST
400	Bad Request — invalid input or business rule violation
401	Unauthorized — missing or expired token
403	Forbidden — valid token but insufficient role
404	Not Found — resource does not exist
409	Conflict — duplicate record
413	Payload Too Large — file upload exceeds size limit
422	Unprocessable Entity — Pydantic validation failure with field details
500	Internal Server Error — unexpected failure

1. Authentication Endpoints

POST /auth/login

Property	Value
Method	POST
Endpoint	/auth/login
Authentication	None (public)
Description	Authenticates a user with email and password. Returns a short-lived access token and a long-lived refresh token. Role is embedded in the JWT payload and verified server-side on every subsequent request.

Request Body:

```
{
```

```
  "email": "string (required)",
  "password": "string (required)"
}
```

Success Response:

```
HTTP 200
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "refresh_token": "dGhpcyBpcyBhIHJlZnJlc2ggdG9rZW4...",
  "token_type": "bearer",
  "expires_in": 900,
  "role": "supervisor"
}
```

Error Responses:

```
401 — { "error": { "code": "INVALID_CREDENTIALS", "message": "Invalid email or password", "status": 401 } }
422 — { "error": { "code": "VALIDATION_ERROR", "message": "email: field required", "status": 422 } }
429 — { "error": { "code": "RATE_LIMIT", "message": "Too many login attempts. Try again in 60 seconds.", "status": 429 } }
```

Example Request:

```
POST /api/v1/auth/login
Content-Type: application/json
{ "email": "meena@site.com", "password": "SecurePass123!" }
```

Example Response:

```
{ "access_token": "eyJ...", "refresh_token": "dGh...", "token_type": "bearer", "expires_in": 900, "role": "supervisor" }
```

POST /auth/refresh

Property	Value
Method	POST
Endpoint	/auth/refresh
Authentication	Refresh token (in body)
Description	Rotates the refresh token and issues a new access token. The old refresh token is invalidated on use.

Request Body:

```
{ "refresh_token": "string (required)" }
```

Success Response:

```
HTTP 200
{ "access_token": "eyJ...", "refresh_token": "new_token...", "expires_in": 900 }
```

Error Responses:

```
401 — { "error": { "code": "TOKEN_EXPIRED", "message": "Refresh token expired or invalid", "status": 401 } }
```

Example Request:

```
POST /api/v1/auth/refresh
{ "refresh_token": "dGh..." }
```

Example Response:

```
{ "access_token": "eyJnew...", "refresh_token": "dGhnew...", "expires_in":
900 }
```

GET /auth/me

Property	Value
Method	GET
Endpoint	/auth/me
Authentication	Bearer token (any role)
Description	Returns the authenticated user's profile. Useful for the frontend to confirm role and display user info after login.

Success Response:

```
HTTP 200
{ "id": "u2", "name": "Meena Rao", "email": "meena@site.com", "role":
"supervisor", "site_id": "s1" }
```

Error Responses:

```
401 — { "error": { "code": "TOKEN_MISSING", "message": "Authorization header
required", "status": 401 } }
```

Example Request:

```
GET /api/v1/auth/me
Authorization: Bearer eyJ...
```

Example Response:

```
{ "id": "u2", "name": "Meena Rao", "email": "meena@site.com", "role":
"supervisor", "site_id": "s1" }
```

---

2. Violations Endpoints

GET /violations

Property	Value
Method	GET
Endpoint	/violations
Authentication	Bearer token (Admin, Supervisor)
Description	Returns a paginated, filterable list of violations. Supervisors see only their own site. Admins see all sites.

Request Parameters:

```
?site_id=uuid          — filter by site (admin only)
?zone_id=uuid          — filter by zone
?violation_type=string — no_helmet | no_vest | restricted_zone_entry
?status=string         — unacknowledged | acknowledged
?from=ISO8601          — start of date range
```

```
?to=ISO8601            — end of date range
?page=int              — page number (default: 1)
?limit=int             — records per page (default: 20, max: 100)
```

Success Response:

```
HTTP 200
{
  "total": 47,
  "page": 1,
  "limit": 20,
  "data": [
    {
      "id": "v1",
      "violation_type": "no_helmet",
      "detected_at": "2024-07-15T08:30:00Z",
      "gate_id": "g1",
      "zone_id": null,
      "worker_id": "u3",
      "screenshot_url": "https://storage.../v1.jpg",
      "confidence_score": 0.923,
      "status": "unacknowledged"
    }
  ]
}
```

Error Responses:

```
401 — Token missing or expired
403 — Worker role not permitted
```

Example Request:

```
GET /api/v1/violations?status=unacknowledged&from=2024-07-01T00:00:00Z
Authorization: Bearer eyJ...
```

Example Response:

```
{ "total": 3, "page": 1, "limit": 20, "data": [ { "id": "v1",
"violation_type": "no_helmet", ... } ] }
```

GET /violations/{id}

Property	Value
Method	GET
Endpoint	/violations/{id}
Authentication	Bearer token (Admin, Supervisor)
Description	Returns full detail for a single violation including screenshot URL, bounding-box metadata, and acknowledgement history.

Request Parameters:

```
id — UUID of the violation (path parameter)
```

Success Response:

```
HTTP 200
{
  "id": "v1",
  "violation_type": "no_helmet",
  "detected_at": "2024-07-15T08:30:00Z",
  "gate_id": "g1",
  "gate_name": "Main Entry Gate",
  "worker_id": "u3",
  "worker_name": "Arjun Das",
  "screenshot_url": "https://storage.../v1.jpg",
  "confidence_score": 0.923,
  "status": "unacknowledged",
  "acknowledged_by": null,
  "acknowledged_at": null
}
```

Error Responses:

```
404 — { "error": { "code": "NOT_FOUND", "message": "Violation not found", "status": 404 } }
```

Example Request:

```
GET /api/v1/violations/v1
Authorization: Bearer eyJ...
```

Example Response:

```
{ "id": "v1", "violation_type": "no_helmet", "confidence_score": 0.923, "status": "unacknowledged", ... }
```

PATCH /violations/{id}/acknowledge

Property	Value
Method	PATCH
Endpoint	/violations/{id}/acknowledge
Authentication	Bearer token (Admin, Supervisor)
Description	Marks a violation as acknowledged. Records which supervisor acknowledged it and when. Cannot be reversed (append-only audit trail).

Request Parameters:

```
id — UUID of the violation (path parameter)
```

Success Response:

```
HTTP 200
{ "id": "v1", "status": "acknowledged", "acknowledged_by": "u2", "acknowledged_at": "2024-07-15T09:00:00Z" }
```

Error Responses:

```
404 — Violation not found
409 — { "error": { "code": "ALREADY_ACKNOWLEDGED", "message": "Violation already acknowledged", "status": 409 } }
```

Example Request:

```
PATCH /api/v1/violations/v1/acknowledge
```

```
Authorization: Bearer eyJ...
```

Example Response:

```
{ "id": "v1", "status": "acknowledged", "acknowledged_by": "u2",
"acknowledged_at": "2024-07-15T09:00:00Z" }
```

---

3. Delivery & Material Endpoints

POST /deliveries

Property	Value
Method	POST
Endpoint	/deliveries
Authentication	Bearer token (Supervisor)
Description	Logs a manual delivery entry including vehicle count, material type, quantity, and unit. The backend compares manual_reported_count against the already-recorded ai_vehicle_count for this gate and time window. If manual count is submitted before AI count is available, status is set to pending_ai_count.

Request Body:

```
{
  "gate_id": "uuid (required)",
  "manual_reported_count": "integer >= 0 (required)",
  "material_type": "cement|sand|bricks|steel|pipes|other (required)",
  "material_quantity": "decimal > 0 (required)",
  "quantity_unit": "bags|tons|cubic_meters|pieces (required)"
}
```

Success Response:

```
HTTP 201
{
  "id": "d1",
  "gate_id": "g1",
  "ai_vehicle_count": 12,
  "manual_reported_count": 9,
  "material_type": "cement",
  "material_quantity": 500.00,
  "quantity_unit": "bags",
  "discrepancy_flag": false,
  "discrepancy_count": 1,
  "status": "logged",
  "logged_at": "2024-07-15T14:00:00Z"
}
```

Error Responses:

```
400 — { "error": { "code": "INVALID_GATE", "message": "Gate not found or
inactive", "status": 400 } }
422 — { "error": { "code": "VALIDATION_ERROR", "message": "material_type must
be one of: cement, sand, bricks, steel, pipes, other", "status": 422 } }
403 — Worker/Admin role — only Supervisor may file delivery logs
```

Example Request:

```
POST /api/v1/deliveries
Authorization: Bearer eyJ...
Content-Type: application/json
{
  "gate_id": "g1",
  "manual_reported_count": 9,
  "material_type": "cement",
  "material_quantity": 500.00,
  "quantity_unit": "bags"
}
```

Example Response:

```
{ "id": "d1", "ai_vehicle_count": 12, "manual_reported_count": 9,
"discrepancy_flag": false, "discrepancy_count": 1, ... }
```

GET /deliveries

Property	Value
Method	GET
Endpoint	/deliveries
Authentication	Bearer token (Admin, Supervisor)
Description	Returns paginated delivery log list. Supervisors see only their own site. Supports filters by gate, date range, and discrepancy status.

Request Parameters:

```
?gate_id=uuid
?from=ISO8601
?to=ISO8601
?discrepancy_flag=boolean
?page=int (default 1)
?limit=int (default 20)
```

Success Response:

```
HTTP 200
{ "total": 15, "page": 1, "data": [ { "id": "d1", "material_type": "cement",
"material_quantity": 500, "quantity_unit": "bags", "ai_vehicle_count": 12,
"manual_reported_count": 9, ... } ] }
```

Error Responses:

```
401 — Token missing
403 — Worker role not permitted
```

Example Request:

```
GET /api/v1/deliveries?gate_id=g1&from=2024-07-01T00:00:00Z
Authorization: Bearer eyJ...
```

Example Response:

```
{ "total": 5, "page": 1, "data": [ { "id": "d1", "material_type": "cement",
... } ] }
```

GET /deliveries/discrepancies

Property	Value
Method	GET
Endpoint	/deliveries/discrepancies
Authentication	Bearer token (Admin ONLY)
Description	Returns delivery logs flagged with repeated count discrepancies, routed exclusively to Admin to avoid conflict of interest with the filing Supervisor. Supervisors receive 403 on this endpoint intentionally.

Request Parameters:

```
?gate_id=uuid
?from=ISO8601
?to=ISO8601
?page=int
```

Success Response:

```
HTTP 200
{
  "total": 3,
  "data": [
    {
      "id": "d2",
      "gate_id": "g1",
      "gate_name": "Main Entry Gate",
      "reported_by": "u2",
      "supervisor_name": "Meena Rao",
      "ai_vehicle_count": 12,
      "manual_reported_count": 9,
      "delta": 3,
      "material_type": "cement",
      "material_quantity": 500.00,
      "quantity_unit": "bags",
      "discrepancy_count": 3,
      "logged_at": "2024-07-15T14:00:00Z"
    }
  ]
}
```

Error Responses:

```
403 — { "error": { "code": "FORBIDDEN", "message": "Only administrators may
view discrepancy escalations", "status": 403 } }
```

Example Request:

```
GET /api/v1/deliveries/discrepancies
Authorization: Bearer eyJ_admin_token...
```

Example Response:

```
{ "total": 1, "data": [ { "id": "d2", "delta": 3, "supervisor_name": "Meena
Rao", "material_type": "cement", ... } ] }
```

GET /deliveries/materials

Property	Value
Method	GET
Endpoint	/deliveries/materials
Authentication	Bearer token (Admin, Supervisor)
Description	Returns material inflow log aggregated by material type over a date range. Used for the site manager material-tracking dashboard view — shows what was delivered, when, and how much.

Request Parameters:

```
?from=ISO8601 (required)
?to=ISO8601 (required)
?gate_id=uuid (optional)
?material_type=string (optional filter)
```

Success Response:

```
HTTP 200
{
  "period": { "from": "2024-07-01", "to": "2024-07-15" },
  "summary": [
    { "material_type": "cement", "total_quantity": 2500.00, "unit": "bags",
"delivery_count": 5 },
    { "material_type": "sand",   "total_quantity": 120.00,  "unit": "tons",
"delivery_count": 3 }
  ],
  "logs": [
    { "id": "d1", "logged_at": "2024-07-15T14:00:00Z", "gate": "Main Gate",
"material_type": "cement", "material_quantity": 500.00, "quantity_unit":
"bags", "logged_by": "Meena Rao" }
  ]
}
```

Error Responses:

```
400 — from and to parameters are required
403 — Worker role not permitted
```

Example Request:

```
GET /api/v1/deliveries/materials?from=2024-07-01T00:00:00Z&to=2024-07-
15T23:59:59Z
Authorization: Bearer eyJ...
```

Example Response:

```
{ "summary": [ { "material_type": "cement", "total_quantity": 2500, "unit":
"bags", "delivery_count": 5 } ], "logs": [ ... ] }
```

---

4. Hazard Report Endpoints

POST /hazard-reports

Property	Value
Method	POST
Endpoint	/hazard-reports
Authentication	Bearer token (Worker)
Description	Submits a hazard report with an optional image upload. The AI categorization service assigns a category asynchronously — the response immediately returns the report ID, and the category is updated within a few seconds. Use GET /hazard-reports/{id} to check the assigned category.

Request Body:

```
Content-Type: multipart/form-data
description: string (required)
location: string (required) — e.g. "North scaffolding, Level 3"
image: file (optional) — JPG or PNG, max 10MB
```

Success Response:

```
HTTP 201
{
  "id": "h1",
  "description": "Loose scaffold plank on Level 3 north side",
  "location": "North scaffolding, Level 3",
  "image_url": "https://storage.../h1.jpg",
  "category": "scaffold_damage",
  "ai_confidence": 0.881,
  "status": "open",
  "reported_at": "2024-07-15T10:00:00Z"
}
```

Error Responses:

```
400 — { "error": { "code": "MISSING_FIELDS", "message": "description and
location are required", "status": 400 } }
413 — { "error": { "code": "FILE_TOO_LARGE", "message": "Image must be under
10MB", "status": 413 } }
415 — { "error": { "code": "INVALID_FILE_TYPE", "message": "Only JPG and PNG
images are accepted", "status": 415 } }
```

Example Request:

```
POST /api/v1/hazard-reports
Authorization: Bearer eyJ_worker_token...
Content-Type: multipart/form-data
description=Loose scaffold plank on Level 3 north side
location=North scaffolding, Level 3
image=<file>
```

Example Response:

```
{ "id": "h1", "category": "scaffold_damage", "ai_confidence": 0.881,
"status": "open", "reported_at": "2024-07-15T10:00:00Z" }
```

GET /hazard-reports

Property	Value
Method	GET
Endpoint	/hazard-reports
Authentication	Bearer token (Admin, Supervisor)
Description	Returns paginated list of hazard reports with AI-assigned categories. Workers cannot access this endpoint — they can only submit.

Request Parameters:

```
?category=string — fire_risk | unsafe_wiring | water_leakage |
scaffold_damage | structural_cracks | uncategorized
?status=string — open | acknowledged | resolved
?from=ISO8601
?to=ISO8601
?page=int
```

Success Response:

```
HTTP 200
{ "total": 8, "data": [ { "id": "h1", "category": "scaffold_damage",
"location": "North scaffolding, Level 3", "status": "open", "reported_at":
"2024-07-15T10:00:00Z", "worker_id": "u3" } ] }
```

Error Responses:

```
403 — Worker role not permitted
```

Example Request:

```
GET /api/v1/hazard-reports?category=scaffold_damage&status=open
Authorization: Bearer eyJ...
```

Example Response:

```
{ "total": 2, "data": [ { "id": "h1", "category": "scaffold_damage",
"status": "open", ... } ] }
```

---

5. Dashboard Endpoint

GET /dashboard/summary

Property	Value
Method	GET
Endpoint	/dashboard/summary
Authentication	Bearer token (Admin, Supervisor)
Description	Returns aggregated KPI data for the dashboard. Supervisors receive site-scoped numbers. Admins receive aggregate across all sites.

Request Parameters:

```
?site_id=uuid (Admin only — omit for all-site aggregate)
```

Success Response:

```
HTTP 200
{
  "ppe_compliance_rate": 92.4,
  "live_worker_count": 47,
```

```
  "active_violations": 3,
  "unacknowledged_violations": 2,
  "todays_vehicle_count_ai": 18,
  "todays_deliveries_logged": 15,
  "open_hazard_reports": 4,
  "escalated_discrepancies": 1,
  "todays_material_summary": [
    { "material_type": "cement", "total_quantity": 1500, "unit": "bags" },
    { "material_type": "sand", "total_quantity": 40, "unit": "tons" }
  ],
  "camera_status": [
    { "gate_id": "g1", "name": "Main Gate", "status": "online", "last_ping":
"2024-07-15T14:59:55Z" },
    { "gate_id": "g2", "name": "South Gate", "status": "offline",
"last_ping": "2024-07-15T13:00:00Z" }
  ]
}
```

Error Responses:

```
401 — Token missing
403 — Worker role not permitted
```

Example Request:

```
GET /api/v1/dashboard/summary
Authorization: Bearer eyJ...
```

Example Response:

```
{ "ppe_compliance_rate": 92.4, "active_violations": 3,
"todays_material_summary": [ { "material_type": "cement", "total_quantity":
1500 } ], "camera_status": [ ... ] }
```

---

6. Equipment Endpoints

GET /equipment

Property	Value
Method	GET
Endpoint	/equipment
Authentication	Bearer token (Admin, Supervisor)
Description	Returns current equipment status list — type, location last seen, idle time, and operational status. Updated by AI detection pipeline.

Request Parameters:

```
?site_id=uuid
?status=active|idle|offline
?equipment_type=string
```

Success Response:

```
HTTP 200
```

```
{ "data": [ { "id": "e1", "equipment_type": "excavator", "identifier": "EXC-001", "last_location": "Zone B", "idle_minutes": 0, "status": "active", "last_seen_at": "2024-07-15T14:58:00Z" } ] }
```

Error Responses:

```
403 — Worker role not permitted
```

Example Request:

```
GET /api/v1/equipment?status=idle
Authorization: Bearer eyJ...
```

Example Response:

```
{ "data": [ { "id": "e1", "equipment_type": "excavator", "status": "idle", "idle_minutes": 45 } ] }
```

---

7. Zone Management Endpoints

POST /zones

Property	Value
Method	POST
Endpoint	/zones
Authentication	Bearer token (Admin ONLY)
Description	Creates a new restricted polygon zone on the site. polygon_coords must be a valid JSON array of x/y coordinate pairs. The backend validates that the polygon is closed (first and last point are the same).

Request Body:

```
{
  "site_id": "uuid (required)",
  "name": "string (required)",
  "zone_type": "crane|high_voltage|excavation|chemical_storage (required)",
  "polygon_coords": [ {"x": 100, "y": 200}, {"x": 300, "y": 200}, {"x": 300, "y": 400}, {"x": 100, "y": 200} ]
}
```

Success Response:

```
HTTP 201
{ "id": "z1", "name": "Crane Zone North", "zone_type": "crane", "polygon_coords": [...], "created_at": "2024-07-15T08:00:00Z" }
```

Error Responses:

```
400 — { "error": { "code": "INVALID_POLYGON", "message": "polygon_coords must form a closed polygon", "status": 400 } }
403 — Only Admin may create zones
```

Example Request:

```
POST /api/v1/zones
Authorization: Bearer eyJ_admin_token...
{ "name": "Crane Zone North", "zone_type": "crane", "polygon_coords": [...] }
```

Example Response:

```
{ "id": "z1", "name": "Crane Zone North", "zone_type": "crane", "created_at":
"2024-07-15T08:00:00Z" }
```

GET /zones

Property	Value
Method	GET
Endpoint	/zones
Authentication	Bearer token (Admin, Supervisor)
Description	Lists all defined zones for a site.

Request Parameters:

```
?site_id=uuid
```

Success Response:

```
HTTP 200
{ "data": [ { "id": "z1", "name": "Crane Zone North", "zone_type": "crane",
"polygon_coords": [...] } ] }
```

Error Responses:

```
403 — Worker not permitted
```

Example Request:

```
GET /api/v1/zones?site_id=s1
Authorization: Bearer eyJ...
```

Example Response:

```
{ "data": [ { "id": "z1", "name": "Crane Zone North", "zone_type": "crane" }
] }
```

---

8. User Management Endpoints

POST /users

Property	Value
Method	POST
Endpoint	/users
Authentication	Bearer token (Admin ONLY)
Description	Creates a new user account. Workers and supervisors cannot self-register — all accounts are created by Admin. Password is hashed server-side before storage.

Request Body:

```
{
  "name": "string (required)",
  "email": "string (required, unique)",
  "password": "string (required, min 8 chars)",
  "role": "supervisor|worker (required — cannot create another admin via
API)",
  "site_id": "uuid (required for supervisor/worker)"
}
```

Success Response:

```
HTTP 201
{ "id": "u4", "name": "Ravi Singh", "email": "ravi@site.com", "role": "worker", "site_id": "s1", "created_at": "2024-07-15T09:00:00Z" }
```

Error Responses:

```
409 — { "error": { "code": "EMAIL_EXISTS", "message": "A user with this email already exists", "status": 409 } }
422 — { "error": { "code": "VALIDATION_ERROR", "message": "password: minimum 8 characters required", "status": 422 } }
```

Example Request:

```
POST /api/v1/users
Authorization: Bearer eyJ_admin_token...
{ "name": "Ravi Singh", "email": "ravi@site.com", "password": "SecurePass1!", "role": "worker", "site_id": "s1" }
```

Example Response:

```
{ "id": "u4", "name": "Ravi Singh", "role": "worker", "created_at": "2024-07-15T09:00:00Z" }
```

PATCH /users/{id}/deactivate

Property	Value
Method	PATCH
Endpoint	/users/{id}/deactivate
Authentication	Bearer token (Admin ONLY)
Description	Soft-deletes a user by setting is_active to false. Deactivated users cannot log in. Historical records (violations, delivery logs) referencing this user are preserved.

Request Parameters:

```
id — UUID of the user (path parameter)
```

Success Response:

```
HTTP 200
{ "id": "u4", "is_active": false, "deactivated_at": "2024-07-15T12:00:00Z" }
```

Error Responses:

```
404 — User not found
409 — { "error": { "code": "CANNOT_DEACTIVATE_SELF", "message": "Admin cannot deactivate their own account", "status": 409 } }
```

Example Request:

```
PATCH /api/v1/users/u4/deactivate
Authorization: Bearer eyJ_admin_token...
```

Example Response:

```
{ "id": "u4", "is_active": false }
```
